

# Savar D Sinha

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## EDUCATION

### University of Chicago

*Ph.D. in Computer Science*

Chicago, IL

*Sept 2026 – Present*

### California Institute of Technology

*B.S. in Computer Science and Physics*

Pasadena, CA

*Sept 2022 – June 2026*

## OBJECTIVE

I am a CS PhD student eager to pursue theoretical research in quantum computation and quantum complexity theory, with an interest in topics at the intersection of theoretical computer science and physics. My primary goal is to analyze promising instances of quantum supremacy to ascertain the degree to which quantum computers truly provide advantage over classical approaches.

## RELEVANT GRADUATE-LEVEL COURSEWORK (FROM CALTECH)

### Computer Science:

- Ph/CS 219ABC — Quantum Computation
- CS 152 — Introduction to Cryptography
- CS/IDS 150B — Probability and Algorithms
- CS/IDS 153 — Current Topics in Theoretical Computer Science: Topics in the Intersection of Theoretical CS and Quantum Computing

### Physics:

- Ph 220 — Quantum Learning Theory
- Ph/CS 219ABC — Quantum Computation
- APh/Ph 138A — Quantum Hardware and Techniques
- Ph 135 — Introduction to Condensed Matter
- Ph 125AB — Quantum Mechanics

## TEACHING EXPERIENCE AND OUTREACH

### Teaching Assistant:

- CS 001X — Intermediate Computer Programming
- CS 011 — Computer Language Lab

### YouTube Lectures (VarPi Channel):

- 60+ videos, 1.5k+ subscribers, 2.5k+ hours of watch time; notable lecture series include
- Algorithms (15 lectures)
  - Quantum Mechanics (11 lectures)
  - Quantum Computation (9 lectures)

## RESEARCH EXPERIENCE

**Prof. John Preskill, California Institute of Technology, Division of Physics, Mathematics, and Astronomy** Sep 2025 – May 2026

- Senior thesis jointly advised by Professor Tong on Lindbladian learning
- Developed techniques for learning local and nonlocal Lindbladians
- Designed Lindbladian-learning protocol robust to state-preparation and measurement error

**Prof. Yu Tong, Duke University, Electrical and Computer Engineering** Jan 2025 – Present

- Developed algorithm for quantum Hamiltonian emptiness testing with sample complexity and evolution time  $\mathcal{O}(1/\epsilon^2)$
- Improved sampling complexity of  $M$ -sparse Hamiltonian learning from  $\mathcal{O}(M^2 \log M)$  to  $\mathcal{O}(M \log M)$
- Developed algorithm for Hamiltonian sparsity testing that achieves evolution time  $\tilde{\mathcal{O}}(M^{1.5}/\epsilon + M/\epsilon^2)$

**Dr. Nathanan Tantivasadakarn, California Institute of Technology, The Division of Physics, Mathematics, and Astronomy** Jun 2024 – Apr 2025

- Developed algorithm for efficient simulation of  $d = 4$  qudit measurements in the stabilizer formalism
- Constructed cluster model to simulate the  $\mathbb{Z}_4$  projective transverse-field Ising model (PTIM) in  $\mathcal{O}(n)$  time
- Demonstrated equivalence between the topological phase transitions in the  $\mathbb{Z}_4$  and coupled  $\mathbb{Z}_2 \times \mathbb{Z}_2$  PTIMs

**Dr. Adel Chaudhuri, Washington University School of Medicine in St. Louis, Department of Radiation Oncology** Jun 2023 – Oct 2023

- Optimized Griffin pipeline to analyze nucleosome protection of prostate cancer marker genes
- Performed Kaplan-Meier and other statistical analyses via R to compare survival rates for risk-stratified patients
- Developed pipeline to calculate conversion rate of lambda phage DNA for genome-wide methylation analysis

**Prof. Jeremy G. Wideman, Arizona State University, Center for Mechanisms of Evolution, Biodesign Institute, School of Life Sciences** Feb 2020 – May 2023

- Identified orthologues of animal and fungal ATP synthase subunits across over 200 eukaryotes
- Utilized bioinformatic tools (BLAST, MUSCLE, and HMMER) and ML-based protein-folding (AlphaFold) on ASU's AGAVE cluster computing server

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#### PUBLICATIONS AND MANUSCRIPTS (6 PUBLISHED, 1 PREPRINT, 80+ CITATIONS)

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- [7] **Sinha SD** (2026) Efficient and SPAM-Robust Ansatz-Free Lindbladian Learning. CaltechTHESIS (Senior Thesis). doi: [10.7907/hprf-fq68](https://doi.org/10.7907/hprf-fq68)
- [6] **Sinha SD** and Tong Y (2025) Improved Hamiltonian learning and sparsity testing through Bell sampling. arXiv (preprint). doi: [10.48550/arXiv.2509.07937](https://doi.org/10.48550/arXiv.2509.07937)
- [5] Chauhan PS, Alahi I, **Sinha SD**, et. al. (2024) Genomic and Epigenomic Analysis of Plasma Cell-Free DNA Identifies Stemness Features Associated with Worse Survival in Lethal Prostate Cancer. Clin Cancer Res. doi: [10.1158/1078-0432.CCR-24-1658](https://doi.org/10.1158/1078-0432.CCR-24-1658)
- [4] **Sinha SD**, Wideman JG. (2023) The persistent homology of mitochondrial ATP synthases. IScience, 26(5), 106700. doi: [10.1016/j.isci.2023.106700](https://doi.org/10.1016/j.isci.2023.106700).
- [3] Macher JN, Coots NL, Girard EB, Langerak A, Muñoz-Gómez SA, **Sinha SD**, Vos R, Wissels R, Gile G, Renema W, Wideman JG. (2023) Single-Cell Genomics Reveals the Divergent Mitochondrial Genomes of Retaria (Foraminifera and Radiolaria). MBio, 0(0), e00302-23. doi: [10.1128/mbio.00302-23](https://doi.org/10.1128/mbio.00302-23)
- [2] Singh RP, Poh Y-P, **Sinha SD**, Wideman JG. (2023) Evolutionary history of oxysterol binding proteins (OSBPs) reveals complex history of duplication and loss in animals and fungi. Contact. 2023;6. doi: [10.1177/25152564221150428](https://doi.org/10.1177/25152564221150428).
- [1] Záhonová K, Lax G, **Sinha SD**, Leonard G, Richards TA, Lukeš J, Wideman JG. (2021) Single-cell genomics unveils a canonical origin of the diverse mitochondrial genomes of euglenozoans. BMC Biology. 19: 103. doi: [10.1186/s12915-021-01035-y](https://doi.org/10.1186/s12915-021-01035-y).

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#### ORAL PRESENTATIONS (7 TOTAL)

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- [8] **Sinha SD** (2026), Efficient and SPAM-Robust Ansatz-Free Lindbladian Learning. Senior Thesis Defense (Pasadena, CA)
- [6, 7] **Sinha SD** et. al (2024 – 2025), Exploring Entanglement Transitions in the 4-Level Qudit Projective Transverse Field Ising Model. Presented at Southern California Conference for Undergraduate Research 2024 (Cal State San Bernadino), National Conference for Undergraduate Research 2025 (Pittsburgh, PA)
- [3 – 5] **Sinha SD** et. al (2023) Combinatorial genomic and epigenomic analysis of plasma cell-free DNA identifies stemness features associated with worse prognosis in high-risk metastatic castration-resistant prostate cancer. Presented at RadOnc Research Retreat (WashU School of Medicine), Summer Undergraduate Research Fellowship 2023 (Caltech), Southern California Conference for Undergraduate Research 2023 (Cal State Fullerton).
- [1, 2] **Sinha SD**, Wideman JG (2020 – 2021) Evolution of Eukaryotic ATP Synthase. Presented at ASU Evolutionary Biology Symposium 2020, ASU CME Lab Meeting Presentation.

## POSTER PRESENTATIONS (3 TOTAL)

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- [3] Hu H, Ma M, Gong W, Ye Q, **Sinha SD**, Tong Y, Flammia S, Yelin S (2026). Heisenberg-limited ansatz-free Hamiltonian learning and sparsity testing. Quantum Information Processing (QIP) Conference. (**Abstract Accepted**).
- [2] Záhonová K, Lax G, **Sinha SD** et al. (2021) Single-cell genomics unveils a canonical origin of the diverse mitochondrial genomes of euglenozoans. International Society of Protistologists 2021 Online Poster (page 62).
- [1] **Sinha SD**, Muñoz-Gómez SA, Wideman JG (2021) The persistence of homology: Diversity and evolution of eukaryotic ATP synthases. International Society of Protistologists 2021 Online Poster (page 83).

## HONORS, AWARDS, AND CERTIFICATIONS

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|-------------------------------------------------------------------------------|------|
| <b>Google Quantum Fellowship (UChicago)</b>                                   | 2026 |
| <b>Jack E. Froehlich Memorial Award Nomination (Top 5% of Class)</b>          | 2025 |
| <b>IBM Quantum Challenge Achievement</b>                                      | 2024 |
| <b>IBM Variational Algorithm Design Certification</b>                         | 2024 |
| <b>IBM Practical Introduction to Quantum-Safe Cryptography Certification</b>  | 2024 |
| <b>IBM Basics of Quantum Information Certification</b>                        | 2023 |
| <b>2023 Radiation Oncology Research Retreat Best Oral Presentation, WashU</b> | 2023 |
| <b>Summer Undergraduate Research Fellowship Award</b>                         | 2023 |

## TECHNICAL SKILLS

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**Languages:** Python, Java, C, C++, Bash, AWK, R, Matlab, x86-64 Assembly, Wolfram Mathematica, OCaml  
**Developer Tools:** VS Code, PyCharm, IntelliJ, Jupyter Notebook, RStudio, Git  
**Python Libraries:** pandas, NumPy, SciPy, Matplotlib, TensorFlow, sklearn, pytorch, qiskit

## ADDITIONAL ACTIVITIES

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|----------------------------------------------------------------|----------------|
| <b>Caltech Chess Club</b>                                      | 2022 – Present |
| • Played for Caltech in Chess.com Collegiate Chess League      |                |
| <b>Cello</b>                                                   | 2014 – Present |
| • Qualified for Arizona All-State Orchestra (2022)             |                |
| • Qualified for Arizona Regionals Orchestra (2020 – 2022)      |                |
| • Caltech Symphony Orchestra (2022 – present)                  |                |
| • Chandler Symphony Orchestra (2019 – 2022)                    |                |
| • Metropolitan Youth Symphony (2018 – 2019)                    |                |
| • Phoenix Youth Symphony (2017 – 2018)                         |                |
| <b>Hindustani Classical Vocal</b>                              | 2012 – Present |
| • Won 1st Place in Bharathotsav International 2021 competition |                |
| • Sangeetshri School of Music (2018 – 2020)                    |                |
| • Phoenix Gharana (2012 – 2017)                                |                |